

OLG - Saving and Growth

1. How X affect s_t

$$\text{Sub } c_{t+1,2} \Rightarrow V(c_{t+1}) = \log(c_{t+1}) + \beta \log((y_{t+1} - c_{t+1})R + x y_{t+1})$$

$$\frac{\partial V}{\partial c_{t+1}} = \frac{1}{c_{t+1}} + \beta \frac{1}{(y_{t+1} - c_{t+1})R + x y_{t+1}} \cdot \frac{\partial}{\partial c_{t+1}} [(y_{t+1} - c_{t+1})R + x y_{t+1}]$$

$$= \frac{1}{c_{t+1}} + \beta \frac{1}{(y_{t+1} - c_{t+1})R + x y_{t+1}} \cdot -R$$

$$\text{FOC: } \frac{1}{c_{t+1}} = \frac{\beta R}{(y_{t+1} - c_{t+1})R + x y_{t+1}}$$

$$\Rightarrow s_t R + x y_{t+1} = \beta R \cdot c_{t+1}$$

$$s_t = \beta c_{t+1} - R^{-1} x y_{t+1}$$

Ans. $X \uparrow \quad s_t \downarrow$

2. 1) Calculate Δk_{t+1} as func. of y_{t+1}
- 2) Calculate $\delta_t = \Delta k_{t+1} / Y_t = \Delta k_{t+1} / (y_{t+1} + y_{t+2})$
- 3) How δ_t relate to G
- 4) How and why 3) depends on β and X/R

$$1): k_{t+1} = s_t N_t$$

$$N=1 \Rightarrow k_{t+1} = s_t \quad \wedge \quad k_t = s_{t-1}$$

$$\text{from part 1.} \Rightarrow s_t = \beta c_{t+1} - R^{-1} x y_{t+1}$$

$$c_{t+1} = y_{t+1} \frac{R+x}{R(1+\beta)} \Rightarrow s_t = \beta \left(y_{t+1} \frac{R+x}{R(1+\beta)} \right) - R^{-1} x y_{t+1}$$

$$= y_{t+1} / R \left[\frac{\beta(R+x)}{1+\beta} - x \right] \cdot \frac{1}{(1+\beta)}$$

$$= y_{t+1} / R \left[\frac{\beta(R+x) - x(1+\beta)}{(1+\beta)} \right]$$

$$= y_{t+1} / R \left[\frac{\beta R - x}{(1+\beta)} \right]$$

$$\text{Ans } \Delta k_{t+1} = k_{t+1} - k_t$$

$$= s_t - s_{t-1}$$

$$= y_{t+1} \frac{(\beta R - x)}{R(1+\beta)} - y_{t+1,1} \frac{(\beta R - x)}{R(1+\beta)}$$

$$y_{t+1} = G y_{t+1,1} \Rightarrow = (G-1) y_{t+1,1} \frac{\beta R - x}{R(1+\beta)} = (G-1) y_{t+1,1} \frac{\beta R - x}{R(1+\beta)}$$

$$2) \delta_t = \Delta k_{t+1} / Y_t$$

$$= (G-1) y_{t+1,1} \frac{\beta R - x}{R(1+\beta)} \frac{1}{(y_{t+1} + y_{t+2})}$$

$$y_{t+2} = x y_{t+1,1} = x y_{t+1} / G \Rightarrow = (G-1) y_{t+1,1} \frac{\beta R - x}{R(1+\beta)} \frac{1}{(y_{t+1} + x y_{t+1,1} / G)}$$

$$\delta_t = \frac{(G-1)}{G} \frac{y_{t+1,1} (\beta R - x)}{R(1+\beta)} \cdot \frac{1}{y_{t+1} (1 + x/G)}$$

$$= \frac{(G-1)}{G} \frac{1}{\left(\frac{G+x}{G} \right)} \cdot \frac{(\beta R - x)}{R(1+\beta)}$$

$$= \left[\frac{G-1}{G+x} \right] \cdot \frac{\beta R - x}{R(1+\beta)}$$

Ans. $\delta_t = \left(\frac{G-1}{G+x} \right) \cdot \left[\frac{B - \frac{x}{R}}{1+B} \right]$

3) suppose $\beta > \frac{x}{R}$, $G \uparrow \Rightarrow \delta_t \uparrow$
 let $f(G) = (G-1)/(G+x)$
 $= (G+x-x-1)/(G+x)$
 $= 1 - \frac{x+1}{x+G}$
 $G \uparrow \Rightarrow f(G) \uparrow \Rightarrow \delta_t \uparrow$

4)

$B = \frac{x}{R}$, $\delta_t = 0$

$B < \frac{x}{R}$, $G \uparrow \Rightarrow \delta_t \downarrow$

4) 1^o addition to 3), β is patience parameter for how much u for old age

x/R is the 'discounted growth'

$\beta > x/R$, patience > 'Expected income discounted' if

we normalize y_{t+1} s.t. $x y_{t+1}/R = x/R$

$\Leftrightarrow s_t > 0$ save more if more patient

$\Rightarrow$ Growth. $G \uparrow$ will make 'Aggregate saving rate δ_t ' increase.

2^o Similarly, $\beta < x/R$ less patient or expect to earn more would rather borrow $\Rightarrow \delta_t$ negative

3^o $\beta = x/R$, income young just enough for consumption young and the same for old age

In this case, agents won't save or borrow

3. $x = rG$ r constant

for simplicity, $\beta = 1/R$ as suggested

$$\delta_t = \left(\frac{G-1}{G+rG} \right) \cdot \left[\frac{1/R - x/R}{1+B} \right]$$

$$= \left[\frac{1}{1+r} - \frac{1}{G(1+r)} \right] \cdot \left[\frac{1-x}{R(1+B)} \right]$$

$$= \frac{G-1}{G(1+r)} \cdot \frac{1-rG}{R(1+B)}$$

$$= \frac{1}{R(1+B)(1+r)} \cdot \frac{(G-1)(1-rG)}{G}$$

$$\text{Let } f(G) = \frac{(G-1)(1-rG)}{G}$$

$$= (G - rG^2 - 1 + rG) / G$$

$$= 1 - rG - G^{-1} + r$$

$$\frac{df}{dG} = -r + G^{-2}$$

Ans. $\frac{1}{G^2} > r \Rightarrow f'(G) > 0, G \uparrow \Rightarrow G_t \uparrow$

$\frac{1}{G^2} = r \Rightarrow f''(G) = -2G^{-3} < 0 \Rightarrow G_t \text{ global max}$

$\frac{1}{G^2} < r \Rightarrow f'(G) < 0 \Rightarrow G \uparrow \Rightarrow G_t \downarrow$

4. old to young 0.7 income ratio

$y_{gap} \approx 4\% \quad y_{us} = 1\% \quad , \quad B = 1/R$

Ratio = y_{t+2} / y_{t+1}

$= x_{y_{t+1}} / G \quad / y_{t+1}$

$= x / G = 0.7$

Suppose $x = rG$ then old to young ratio = $r = 0.7$

by part 3. for contradiction suppose OLG explain $G \uparrow \Rightarrow G_t \uparrow$

$\Leftrightarrow r < \frac{1}{G^2}$

$0.7 < \frac{1}{G^2}$

$G < \sqrt{1/0.7} \approx 1.195$

suppose generation gap = $55 - 25 = 30$

$G_{gap} = (1+y)^{30} = (1.04)^{30} \approx 3.243 > 1.195$

in this case Δt would be negative contradicts

$\Rightarrow$ OLG cannot explain